

BYTEDANCE SEED

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VAPO: Value-Model-Based RL for Advanced Reasoning

Achieves state-of-the-art on AIME 2024, surpassing DAPO
(50) and DeepSeek-R1-Zero (47) by 10+ points

arxiv:2504.05118 · AIME 2024 avg@32 · 5,000 training steps

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought

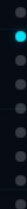
Value-model-free methods (GRPO, DAPO) compute trajectory-level advantage and assign it uniformly. Value models enable precise token-level credit assignment — critical when subtle step-level errors cause catastrophic failures.

Value-Model-Free

- Trajectory-level advantage
- Uniform token assignment
- Higher variance estimates
- Less sample efficient

Value-Model-Based

- Token-level credit assignment
- Precise per-step impact tracing
- Lower-variance estimates
- Better generalization



Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

01 Value Model Bias

Long trajectories and bootstrapped learning introduce severe bias into value estimates, destabilizing early training.

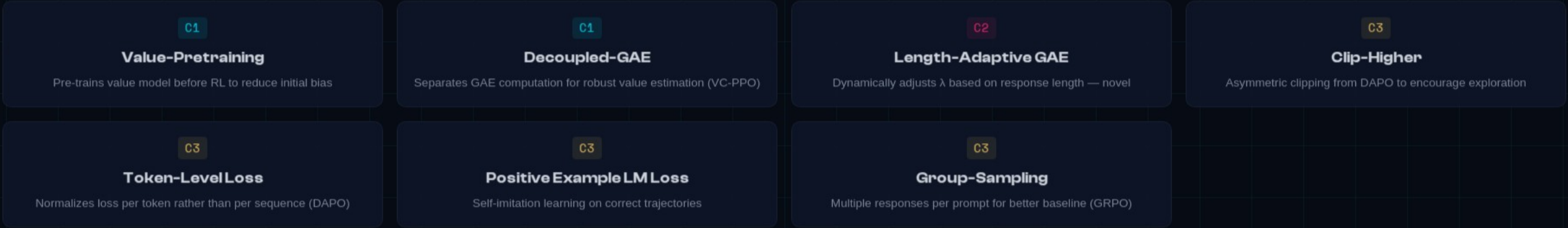
02 Heterogeneous Lengths

Variable response lengths require different bias-variance tradeoffs — a single fixed λ cannot optimize all sequences.

03 Reward Sparsity

Only the final `<eos>` token receives reward, making credit assignment across thousands of tokens extremely hard.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL



■ Challenge 1: Bias ■ Challenge 2: Heterogeneous Lengths ■ Challenge 3: Sparsity

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

A novel technique that dynamically adjusts the λ parameter in GAE based on response length, optimizing the bias-variance tradeoff across heterogeneous sequences.

Short Responses

Lower $\lambda \rightarrow$ lower variance. Short sequences have denser reward signal; we prioritize stable, low-variance estimates.



λ Low

Long Responses

Higher $\lambda \rightarrow$ lower bias. Long sequences need reduced bootstrapping bias; we accept slightly higher variance for accurate credit.



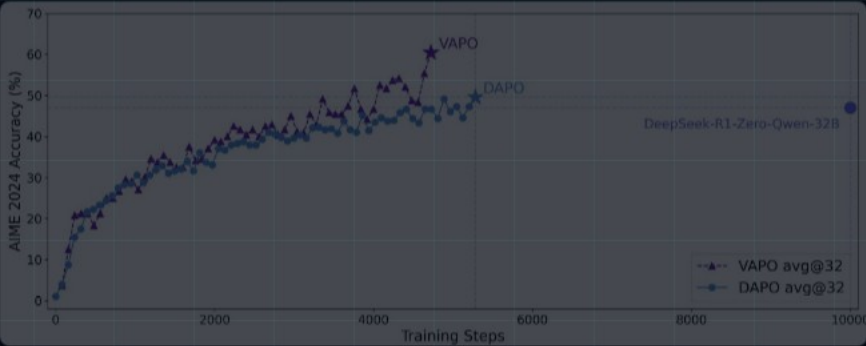
λ High



VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero

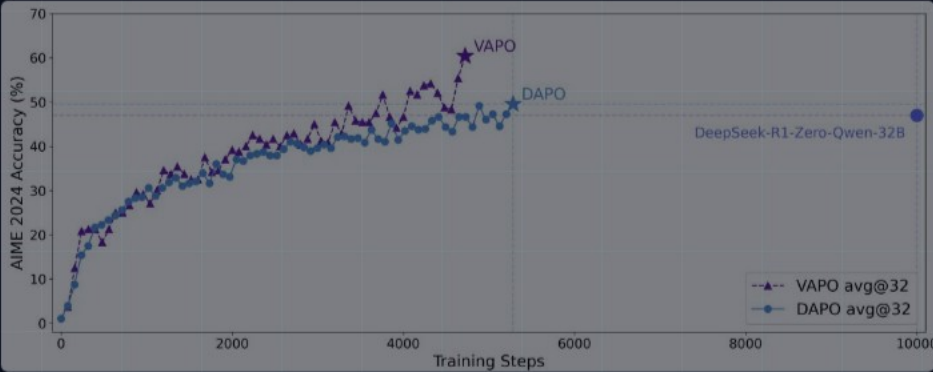


Training curve: VAPO vs DAPO on AIME 2024



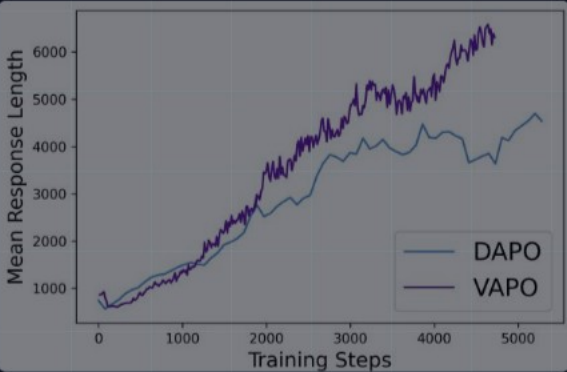
VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

Accuracy Over Training



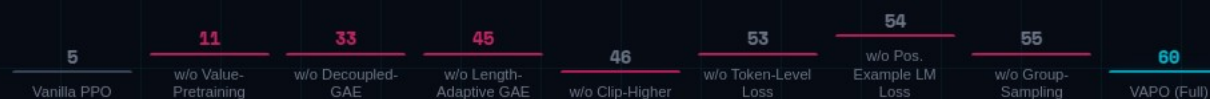
VAPO hits ~60% at step 4,500. DAPO plateaus near 50% at step 5,500. DeepSeek-R1-Zero reaches 47% after ~10,000 steps.

Response Length Over Training



VAPO generates longer reasoning chains (~6,000 tokens vs DAPO's ~4,500), suggesting more elaborate exploration.

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points



Critical insight: Removing Value-Pretraining drops score from 60 → 11. Without it, value-model-based RL barely works. Decoupled-GAE is second most impactful (drop to 33), followed by Length-Adaptive GAE (drop to 45).

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1 Value-model-based can surpass value-free

When bias, heterogeneous lengths, and reward sparsity are systematically addressed, value-based RL outperforms GRPO/DAPO by 10+ points.

2 Length-Adaptive GAE is critical for long CoT

Dynamic λ adjustment uniquely solves the heterogeneous sequence length problem, contributing 15 points to final performance.

3 Value pretraining is the single most important component

Contributing 49 points (11 $\rightarrow$ 60). Without it, value-model-based RL collapses — pretraining eliminates catastrophic initial bias.

4 VAPO is highly stable and efficient

Reaches SOTA in $\sim$ 5,000 steps with zero training crashes across multiple independent runs — a reliability benchmark.

5 Implications for future reasoning models

Value-model-based RL should be revisited for any complex reasoning task with long chain-of-thought and sparse rewards.

